Project PRISM

Multiple Allelism

Genetics and Evolution

What is Multiple Allelism?

- **Multiple allelism** means that a gene has **more than two allelic forms** in a population.
- Even so, each **diploid individual** still carries only **two alleles** for that gene, one inherited from each parent.
- Multiple allelism increases the number of possible **genotypes** in a population.
- It may also increase the number of possible **phenotypes**, depending on how the alleles interact.

Population Level vs Individual Level

- Multiple allelism is a concept at the **population level**, not the individual level.
- A population may contain **three or more alleles** at one locus.
- But one individual can carry only **two** of those alleles.
- Therefore, multiple allelism does **not** change the fact that alleles still occur in pairs in diploid organisms.
- It also does **not** violate the law of segregation.

Why is Multiple Allelism Important?

- It increases **genetic diversity** within a population.
- It creates **more genotype combinations** than a simple two-allele system.
- It helps explain why some traits show **more than two phenotypes**.
- It shows that inheritance can be more complex than the simplest Mendelian examples, even when only **one gene** is involved.

The ABO Blood Group System

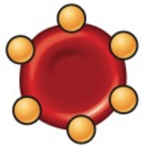
- A classic example of multiple allelism is the **ABO blood group system.**
- The gene has **three alleles**:
  - $\mathbf{I^A}$
  - $\mathbf{I^B}$
  - **i**
- These alleles determine which carbohydrate antigens are present on the surface of red blood cells.
- Because there are three alleles, this single gene produces more genotype combinations than a standard two-allele trait.

▼ **Figure 14.11 Multiple alleles for the ABO blood groups.** The four blood groups result from different combinations of three alleles.

(a) The three alleles for the ABO blood groups and their carbohydrates. Each allele codes for an enzyme that may add a specific carbohydrate (designated by the superscript on the allele and shown as a triangle or circle) to red blood cells.

Allele	I^A	I^B	i
Carbohydrate	A △	B ○	none

(b) Blood group genotypes and phenotypes. There are six possible genotypes, resulting in four different phenotypes.

Genotype	I^AI^A or I^Ai	I^BI^B or I^Bi	I^AI^B	ii
Red blood cell with surface carbohydrates				
Phenotype (blood group)	A	B	AB	O

Dominance Relationships in ABO

- $\mathbf{I^A}$ and $\mathbf{I^B}$ are both dominant over **i**.
- $\mathbf{I^A}$ and $\mathbf{I^B}$ are **codominant** to each other.
- Therefore:

  $\mathbf{I^AI^A}$ or $\mathbf{I^Ai}$ → blood group **A**

  $\mathbf{I^BI^B}$ or $\mathbf{I^Bi}$ → blood group **B**

  $\mathbf{I^AI^B}$ → blood group **AB**

  ii → blood group **O**
- So three alleles give **six possible genotypes** but only **four phenotypes**.

Genotypes and Phenotypes

- The six possible genotypes are:

  I^AI^A

  I^Ai

  I^BI^B

  I^Bi

  I^AI^B

  ii
- These produce four phenotypes:

  A

  B

  AB

  O
- This shows that **different genotypes can give the same phenotype**, such as $\mathbf{I^AI^A}$ and $\mathbf{I^Ai}$ both producing blood group A.

Multiple Allelism vs Codominance

- **Multiple allelism** and **codominance** are different ideas.
- **Multiple allelism** means there are more than two alleles in the population.
- **Codominance** means that two alleles in a heterozygote are both fully expressed.
- In the ABO system:

the locus shows **multiple allelism** because there are three alleles,

and the genotype $\mathbf{I^A I^B}$ shows **codominance** because both antigens are expressed.

How Does Multiple Allelism Arise?

- New alleles arise by **mutation**.
- Over time, repeated mutation in the same gene can generate several allelic forms in a population.
- If these alleles persist, the locus becomes a **multiple-allele system**.
- Therefore, multiple allelism is one consequence of **genetic variation**.
- This connects back to the earlier subtopic on **genetic variation**, where mutation was introduced as the original source of new alleles.

How Does It Still Follow Mendelian Rules?

- Multiple allelism is still consistent with Mendelian inheritance.
- Each individual still inherits:

  one allele from one parent

  and **one allele from the other parent**
- These alleles still **segregate during gamete formation**.
- The only difference is that the **population contains more allele types** than just two.
- So multiple allelism is best understood as an **extension of Mendelian inheritance**, not a contradiction of it.

Biological Importance of ABO

- The ABO phenotype affects the antigens present on red blood cells.
- These antigens are important in **blood transfusion compatibility**.
- Individuals produce antibodies against antigens they do not possess.
- As a result, the genetic basis of ABO blood groups has clear **medical significance**.
- This makes ABO a strong example of how a genotype can affect both **phenotype** and **biological function**.

Additional Pointers

- In IBO, multiple allelism is often tested through:

  genotype-to-phenotype conversion,

  phenotype-to-genotype deduction,

  inheritance prediction in crosses,

  and distinguishing **multiple allelism** from **codominance** or **simple dominance**.
- A common trap is to think that a person can have “three alleles” for one gene; this is false in a diploid organism.
- Another common trap is to confuse “three alleles in the population” with “three phenotypes”; the number of phenotypes depends on the **dominance relationships**, not just the number of alleles.

Conclusion

- Multiple allelism occurs when a gene has **more than two allelic forms** in a population.
- A diploid individual still carries only **two alleles** at that locus.
- The ABO blood group system is the classic example, involving **I^A, I^B, and i**.
- Multiple allelism increases the number of possible genotypes and contributes to phenotypic diversity.
- It is an **extension of Mendelian inheritance** and a result of **genetic variation.**

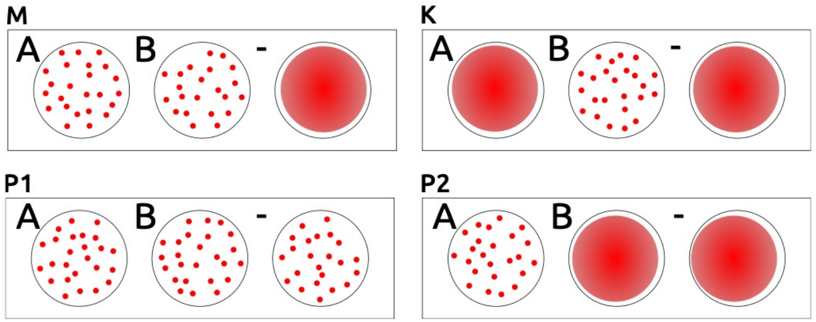
IBO past year questions

IBO2013 Theoretical Exam 2: Qn 19

Three blood groups have been characterized in cats, all of which are encoded by a single gene with three alleles, of which allele A is dominant over allele B, and allele AB is dominant over B, but recessive to A. Most cats with blood groups A or B have anti-B or anti-A antibodies, respectively. Cats with blood group AB do not produce either antibodies anti-A nor anti-B.

Antigens	Produced antibodies		
	Anti-A	Anti-B	Anti-AB
A	-	+	-
B	+	-	-
AB	-	-	-

The figure below shows the results of blood transfusion compatibility tests performed for a mother cat (M), her kitten (K) and two potential father cats (P1 and P2). The cards consist of three circles that contain anti-A (A) and anti-B (B) antibodies, or no antibodies at all as a negative control (-). When adding a drop of blood to the circles, the occurrence of an agglutination reaction becomes visible (red dots).



Indicate if each of the following statements is true or false.

A. Mixing blood of kitten K with serum from P2 should lead to agglutination.

B. M could receive erythrocytes from P2.

C. A back-cross between mother M and kitten K might donate erythrocytes to P2.

D. These results suggest that P1 is the more likely father of K than P2.

Answer Key

A. True **B.** True **C.** False **D.** False

Original commentary
Correct answers
A *true*
P2 produces anti-B, which are present in its serum and would cause an agglutination of erythrocytes of kitten K.
B *true*
The P2 serum has no anti-A and causes therefor no agglutination with antigens of erythrocytes in M.
C *false*
A cat with the blood group A has anti-B, which react both with AB and B antigens on erythrocytes. Since the mother has genotype AB/B and the kitten B/B, any offspring of them has either blood group AB or B.
D *false*
The negative control indicates that the test failed. So the genotype of P1 is unknown and hence these results do not suggest anything. Note, however, that if P1 had indeed blood group AB, he would have the same probability as P2 to be the father of K. The corresponding probability for P2 is either 0% if he had genotype A/B or A/AB or 25% if he had genotype A/B.